

Analysing Political Debates through Multimodal Big Data: Insights from the 2026 Portuguese Presidential Elections

Processing Big Data — Midterm Review

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Part 1

Exploratory Data Analysis & Visualization

Exploratory Data Analysis & Visualization

- **Goal:** Familiarize ourselves with the data
- **Pipeline:**
 - 1 Inspect how each modality (visual, audio, text) is structured per video
 - 2 Check for missing values and assess feature normalization
 - 3 Visualize features for both debates and newscasts
 - 4 Find correlations within modalities

Dataset Overview

Two video sources, multiple modalities per frame/segment

Debate videos (28)

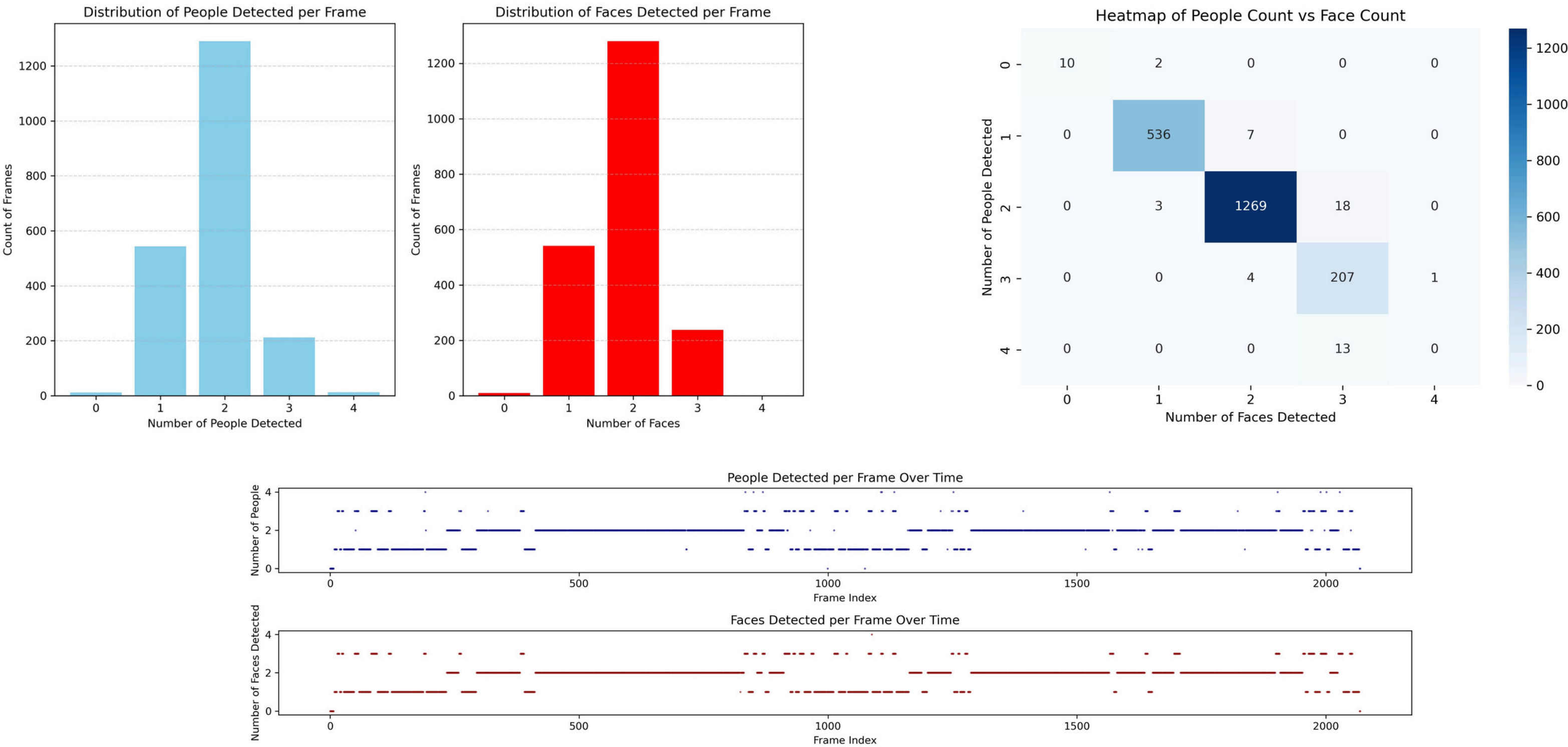
- Visual: poses, faces, emotions, landmarks (1 fps)
- Audio: pitch, HNR, jitter, shimmer, formants
- Speech: pauses, rate, articulation, speaker embeddings (512)
- Transcripts: Whisper + EuroLLM embeddings (4096)
- Text: supporting party, poll timeline

Prime-time newscasts (2 channels)

- Aligned to 7–10 day windows around polls
- Visual: OCR boxes, recognized text, confidence
- Speech: Whisper transcripts + EuroLLM embeddings
- Duration 35 min – 1 h, sampled at 1 fps
- Same days across both channels

Visual Features

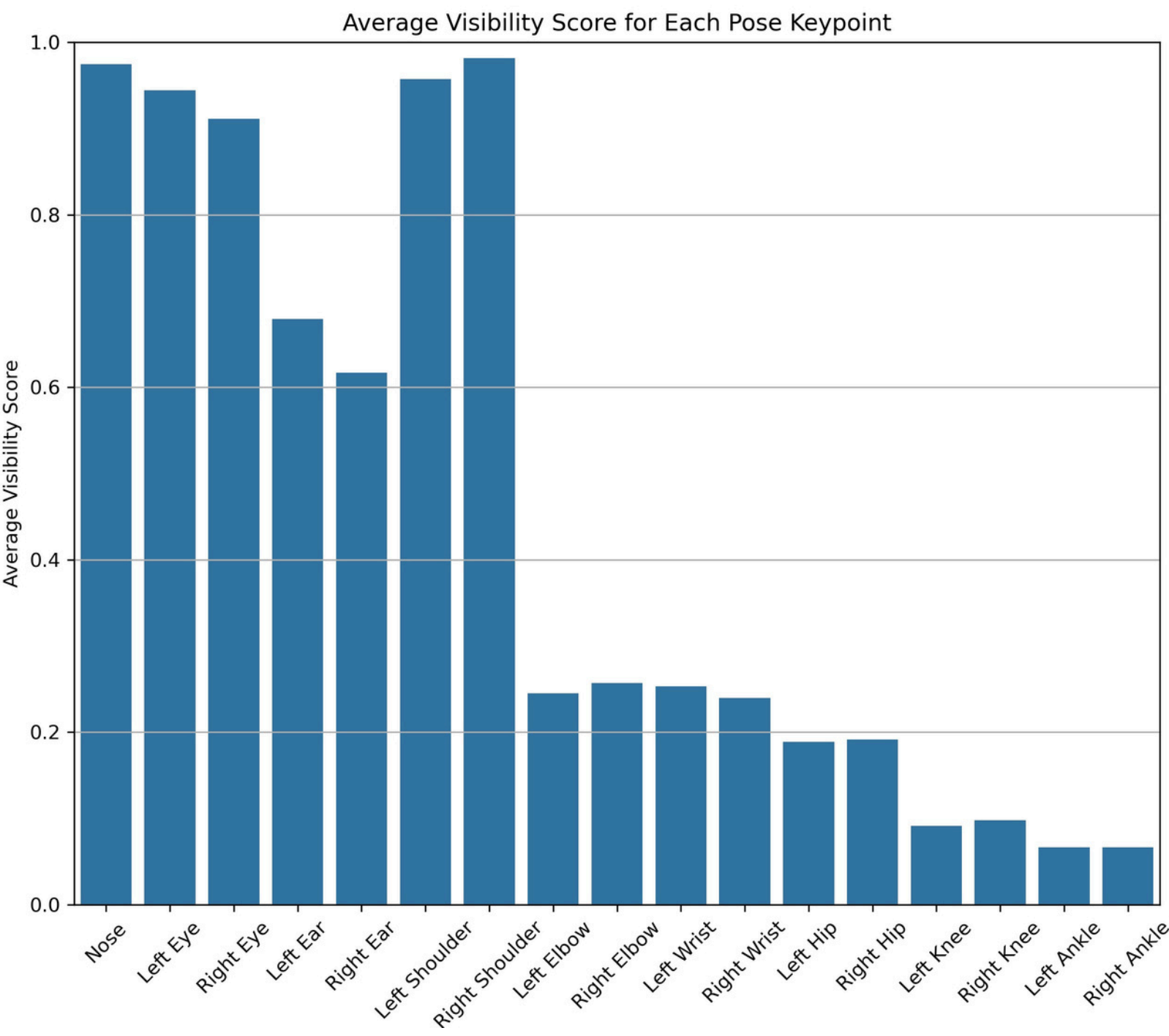
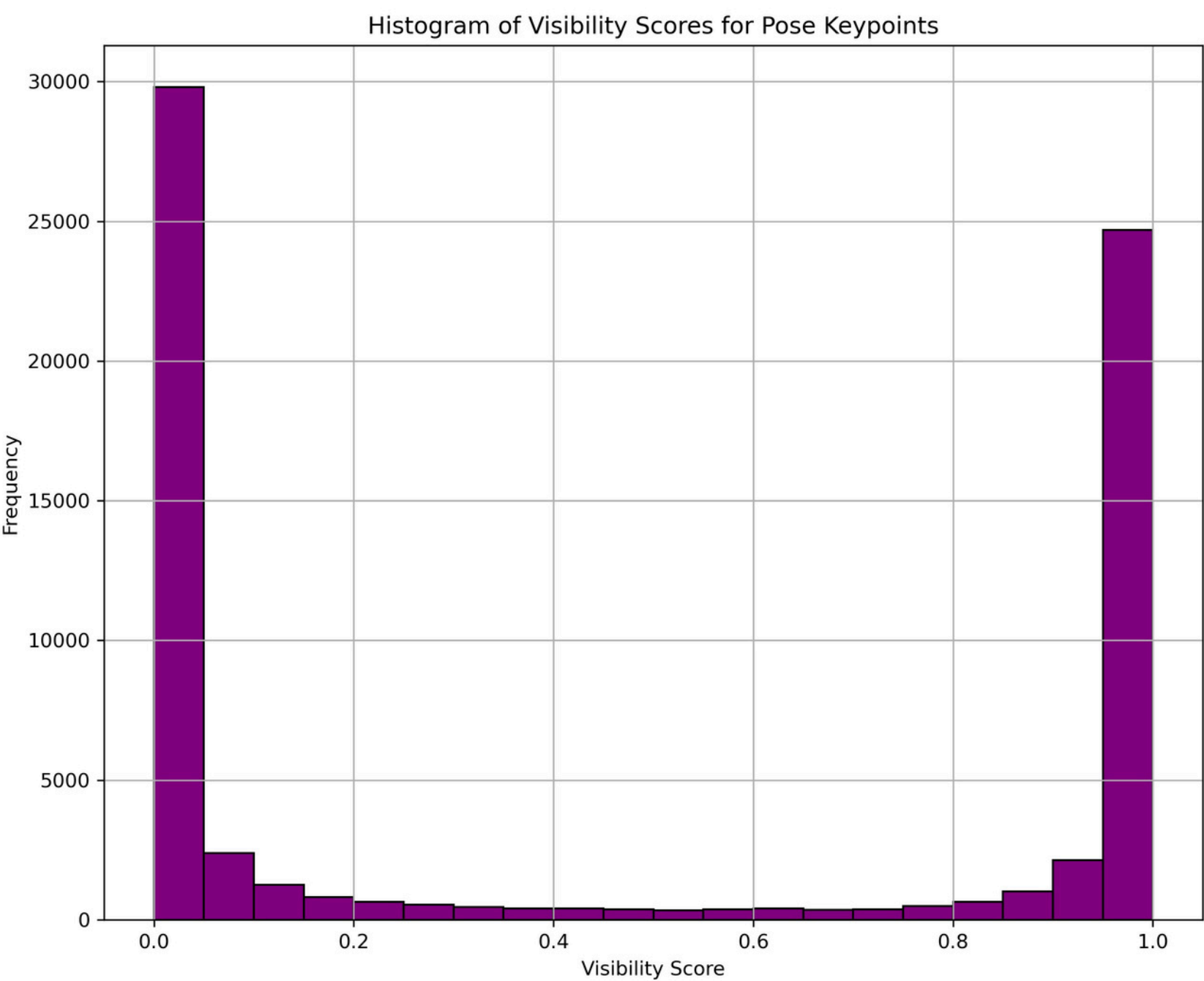
Detection of people and faces, mismatches and detections over time



Example: Seguro vs Ventura

Visual Features

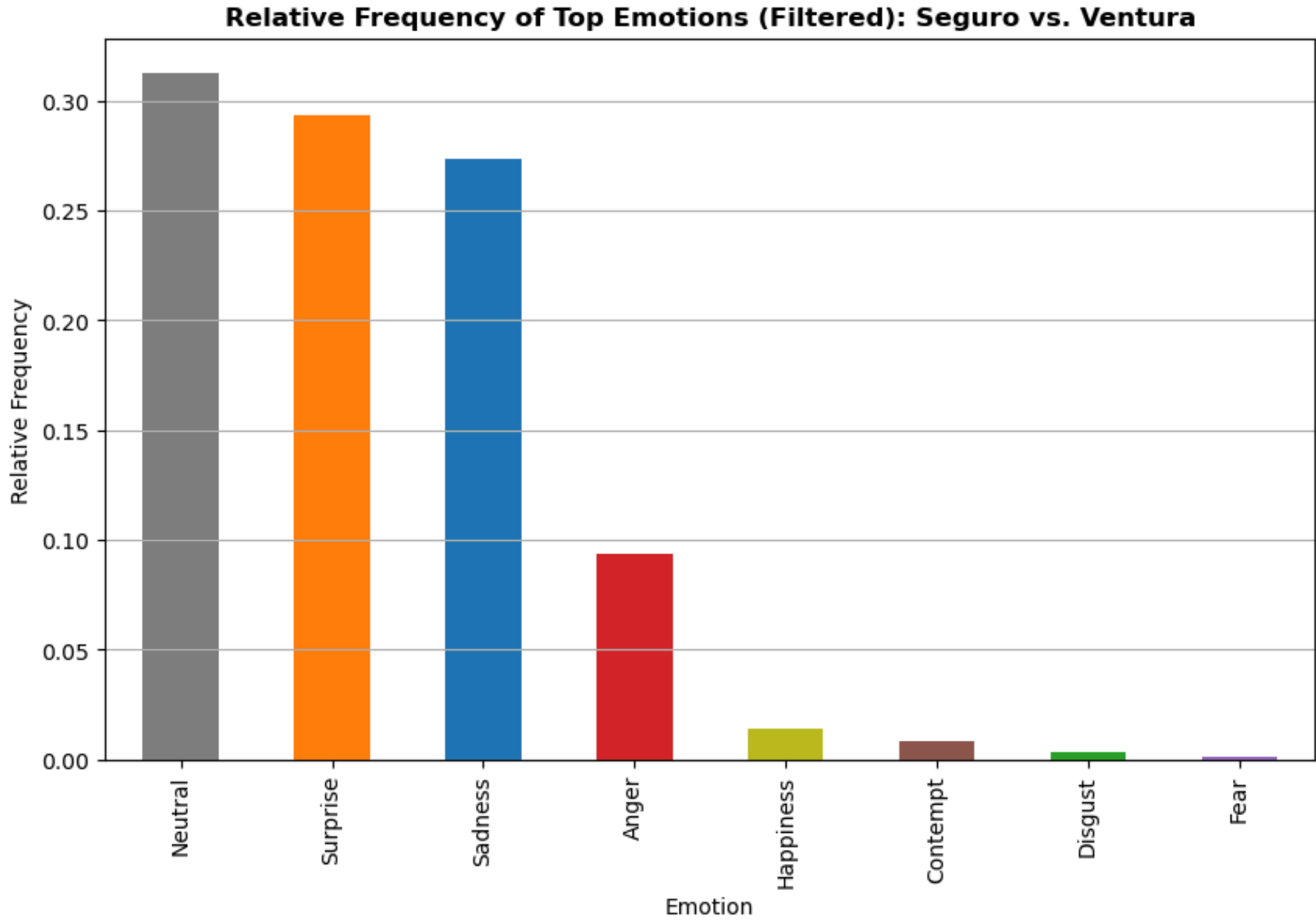
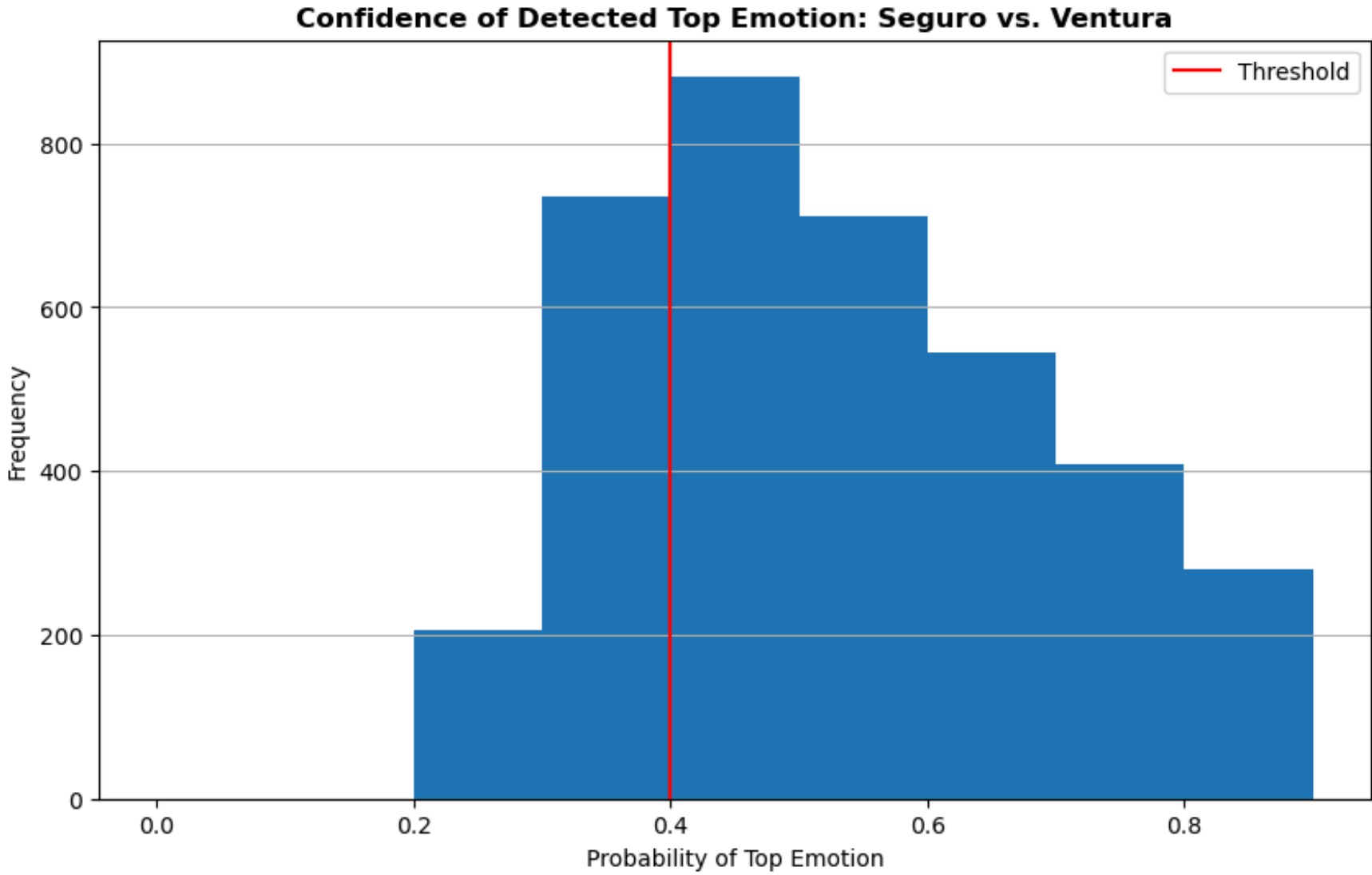
Visibility of pose keypoints in debates



Example: Seguro vs Ventura

Visual Features: Emotions

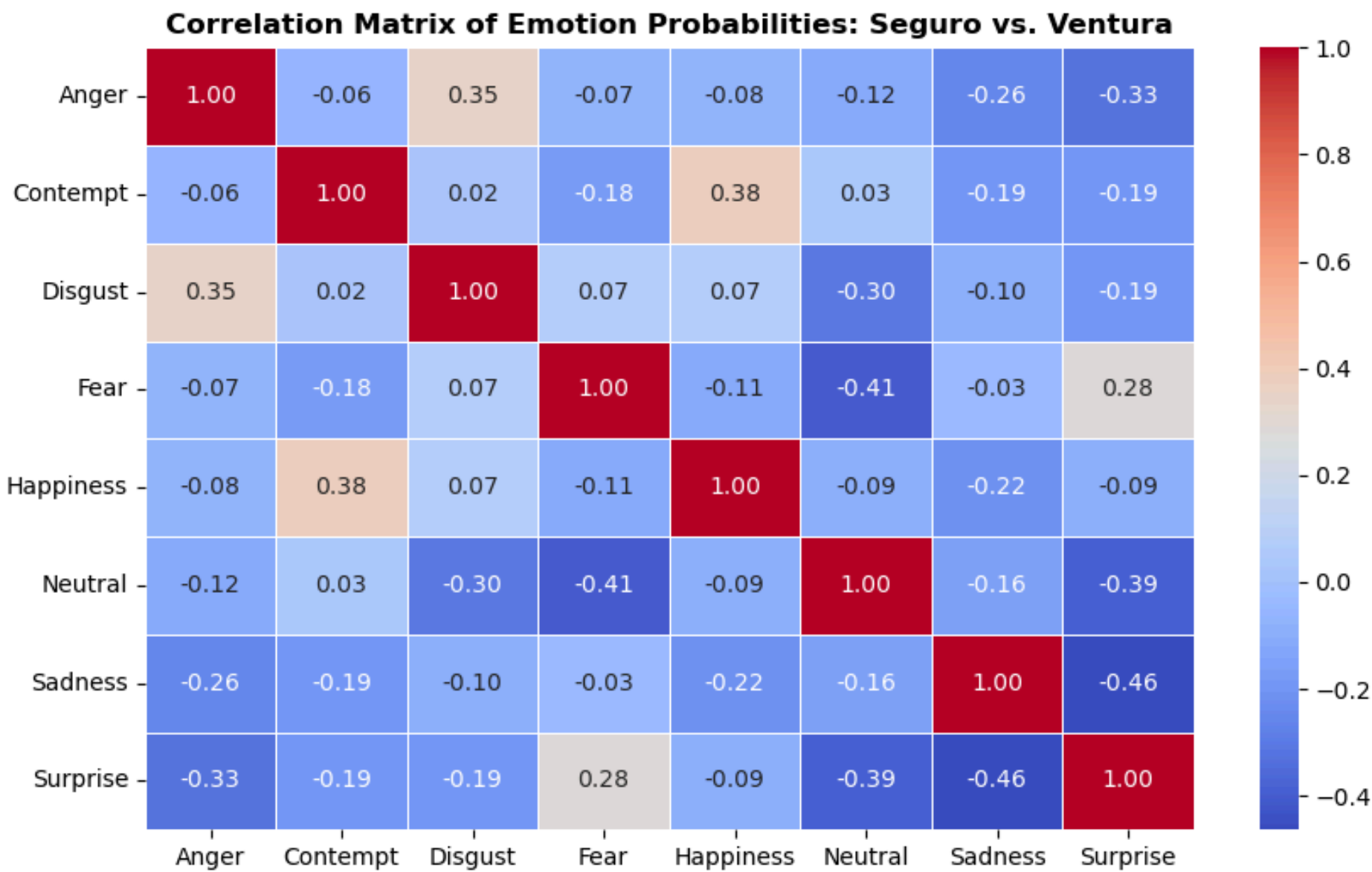
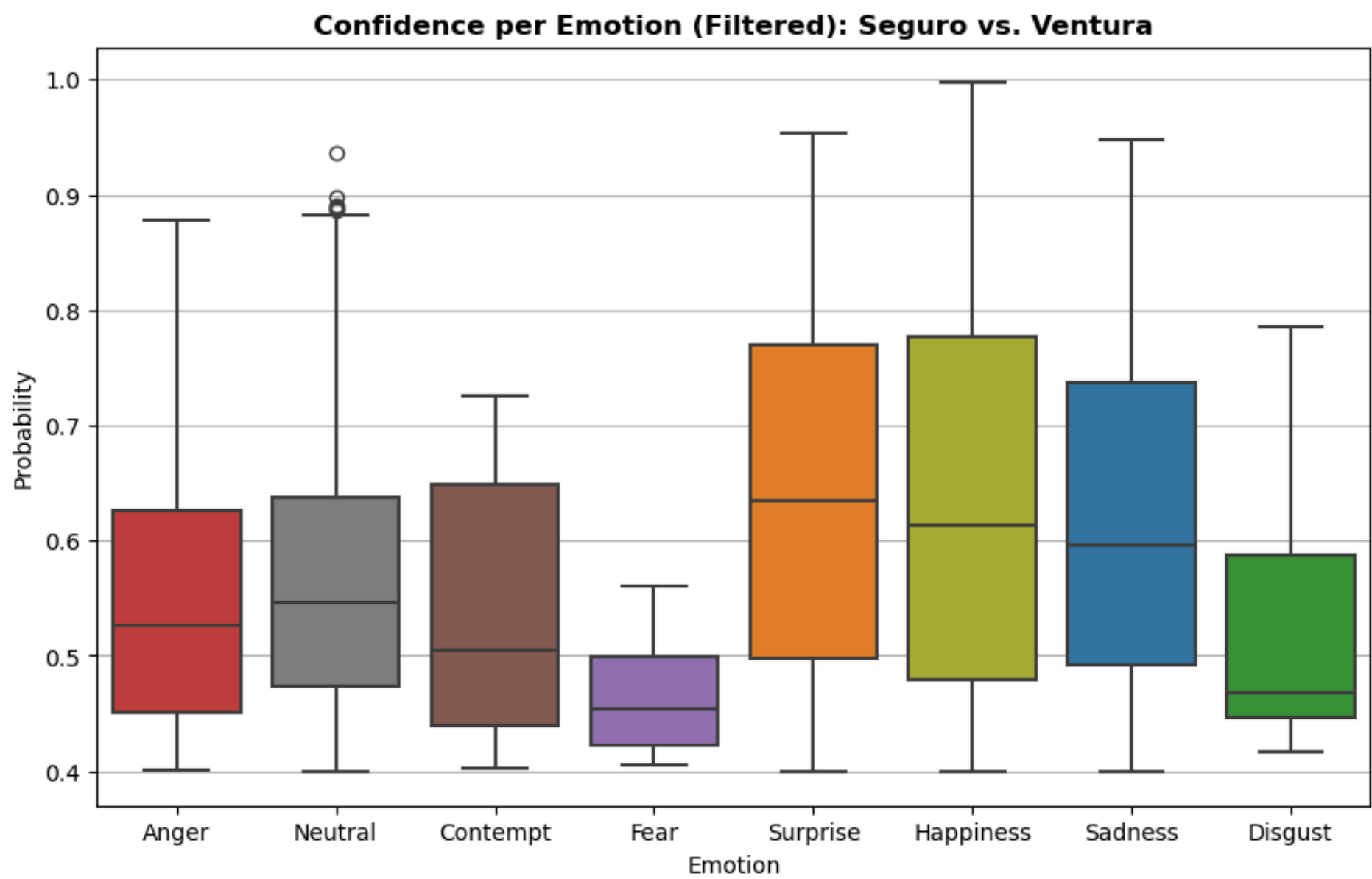
Detected Top Emotions and corresponding model confidence



Example: Seguro vs Ventura

Visual Features: Emotions

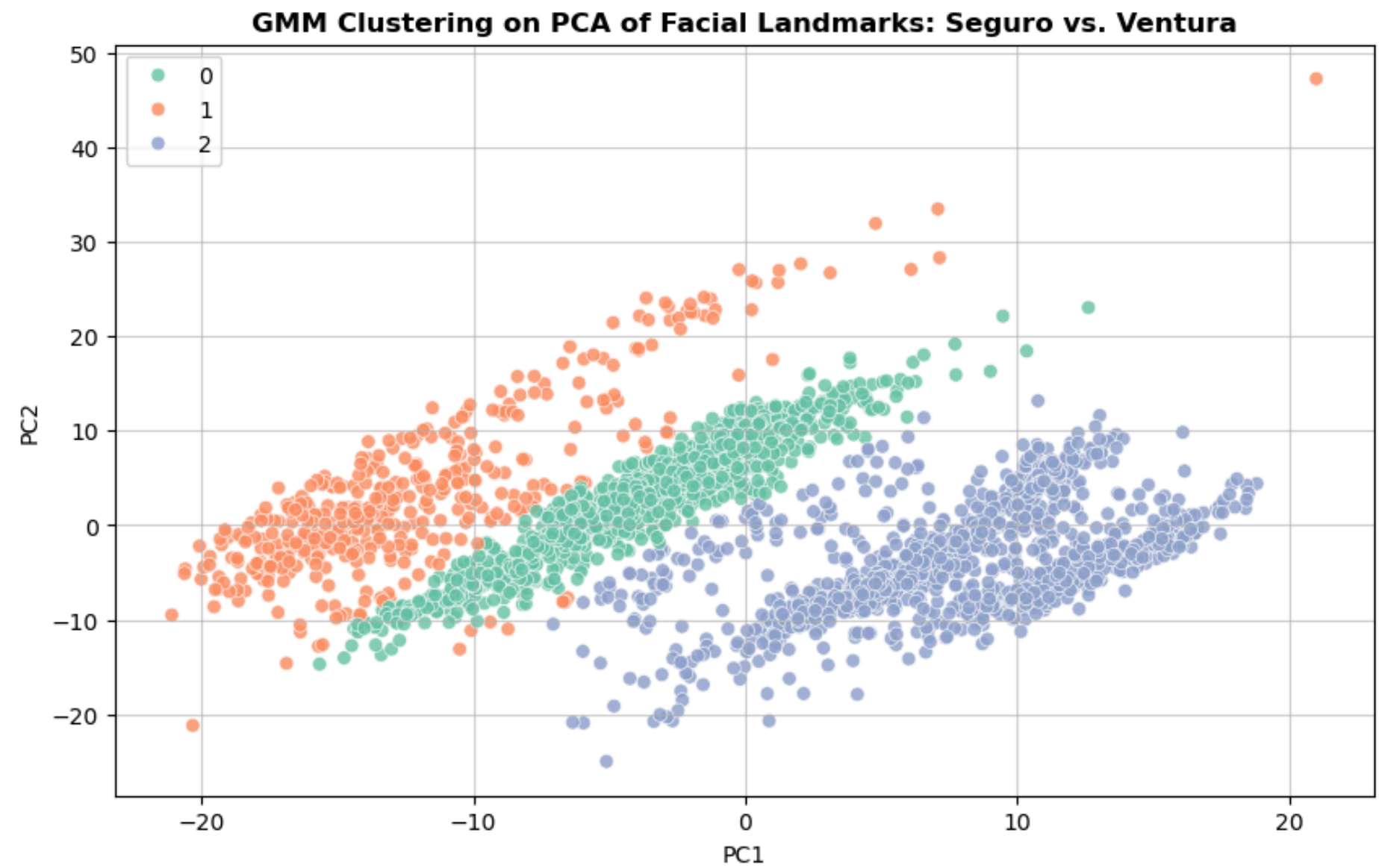
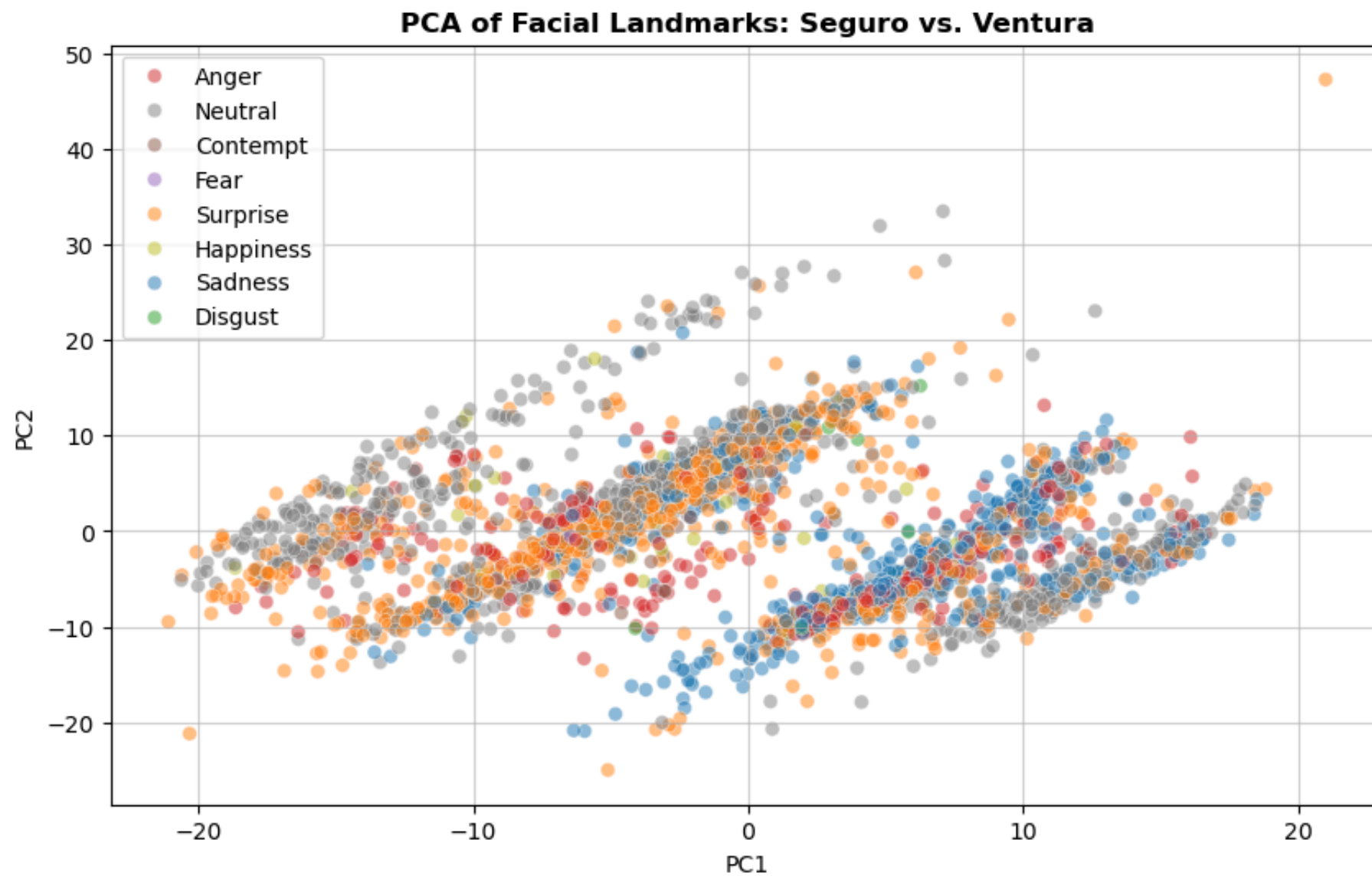
Model's confidence per Emotion and Emotion Correlation Matrix



Example: Seguro vs Ventura

Visual Features

PCA of Facial Landmarks and GMM Clustering



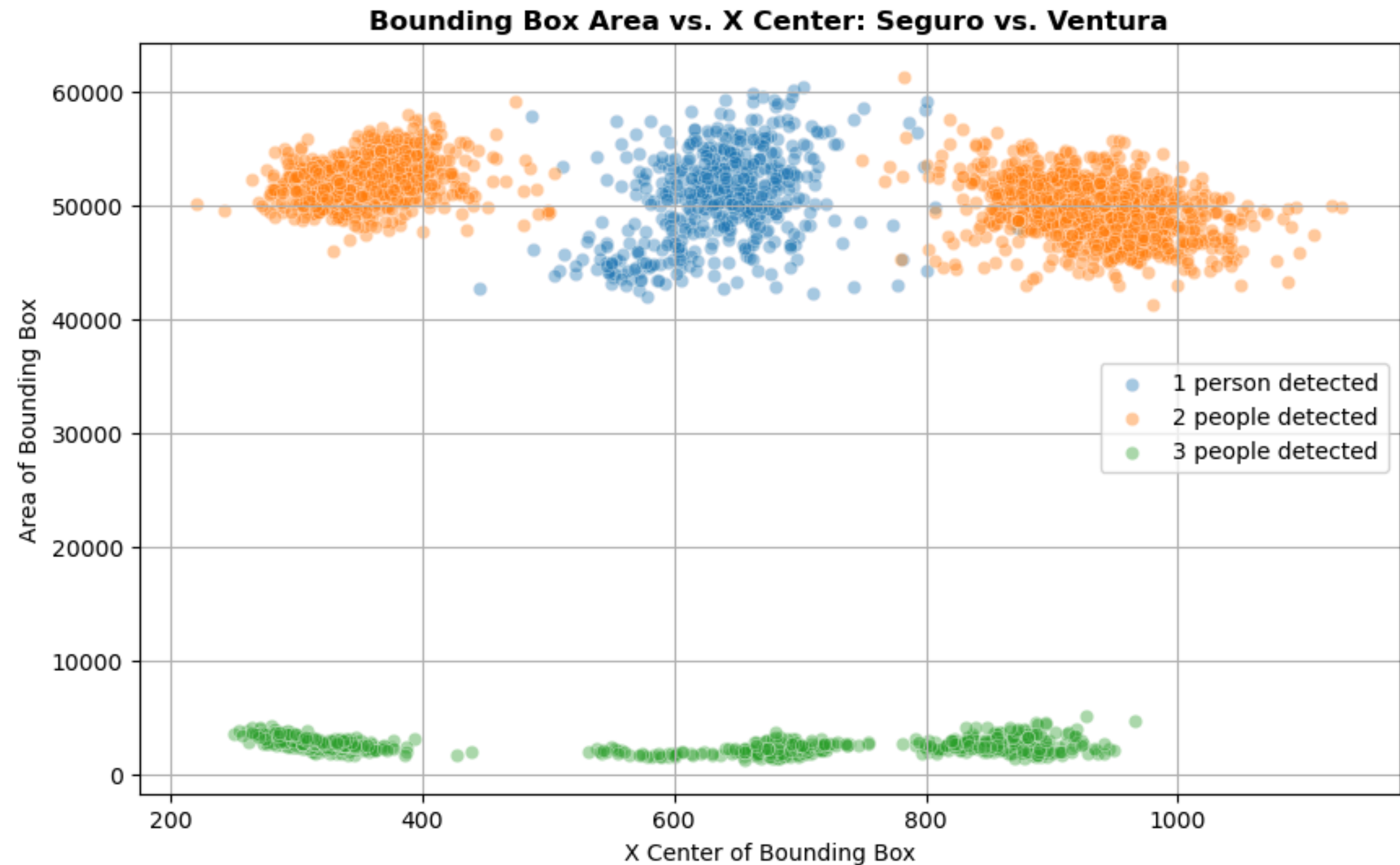
Example: Seguro vs Ventura

Visual Features

Area vs. x-center of each detected face bounding box

Considerations:

- No gestural language interpreters



Example: Seguro vs Ventura

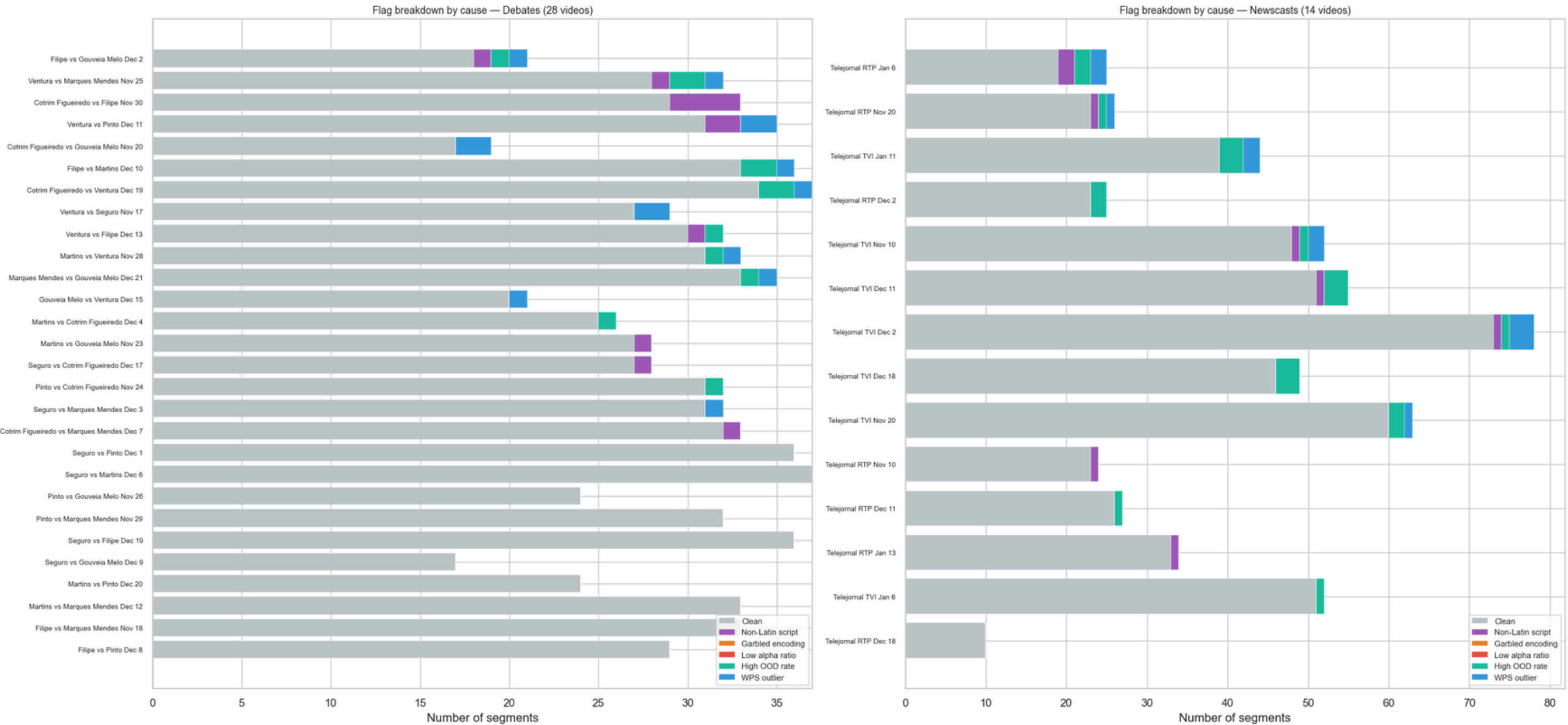
Audio and Speech Features

Audio Transcript Quality (Debates & Newscasts) - Random characters & Non Portuguese Text

Suspicious segment Detection

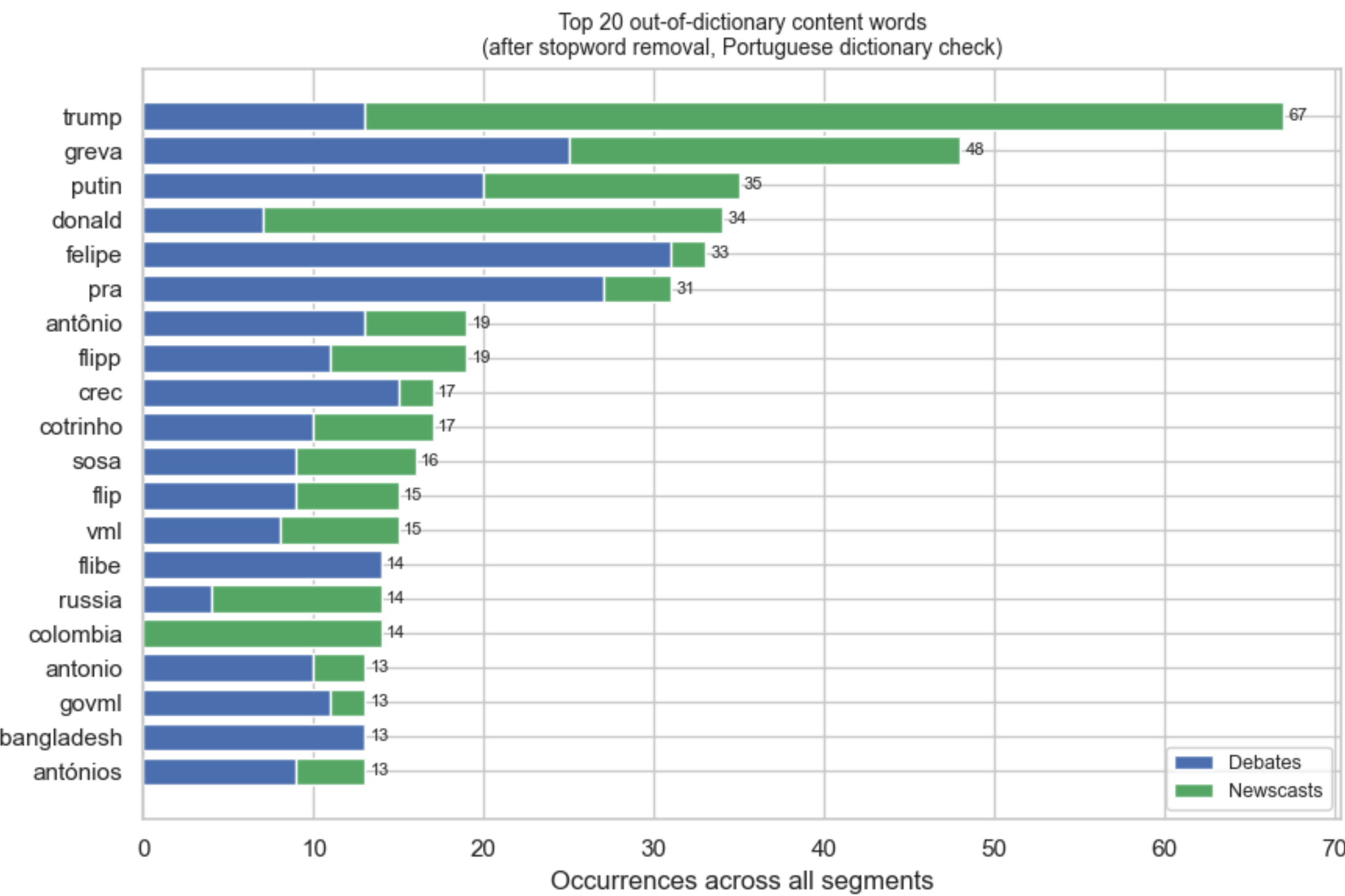
- Low alphabetic ratio
- Abnormal speech rate
- Garbled encoding
- Non-Latin script
- High out-of-dictionary rate

Transcript quality: segment-level flag breakdown per video



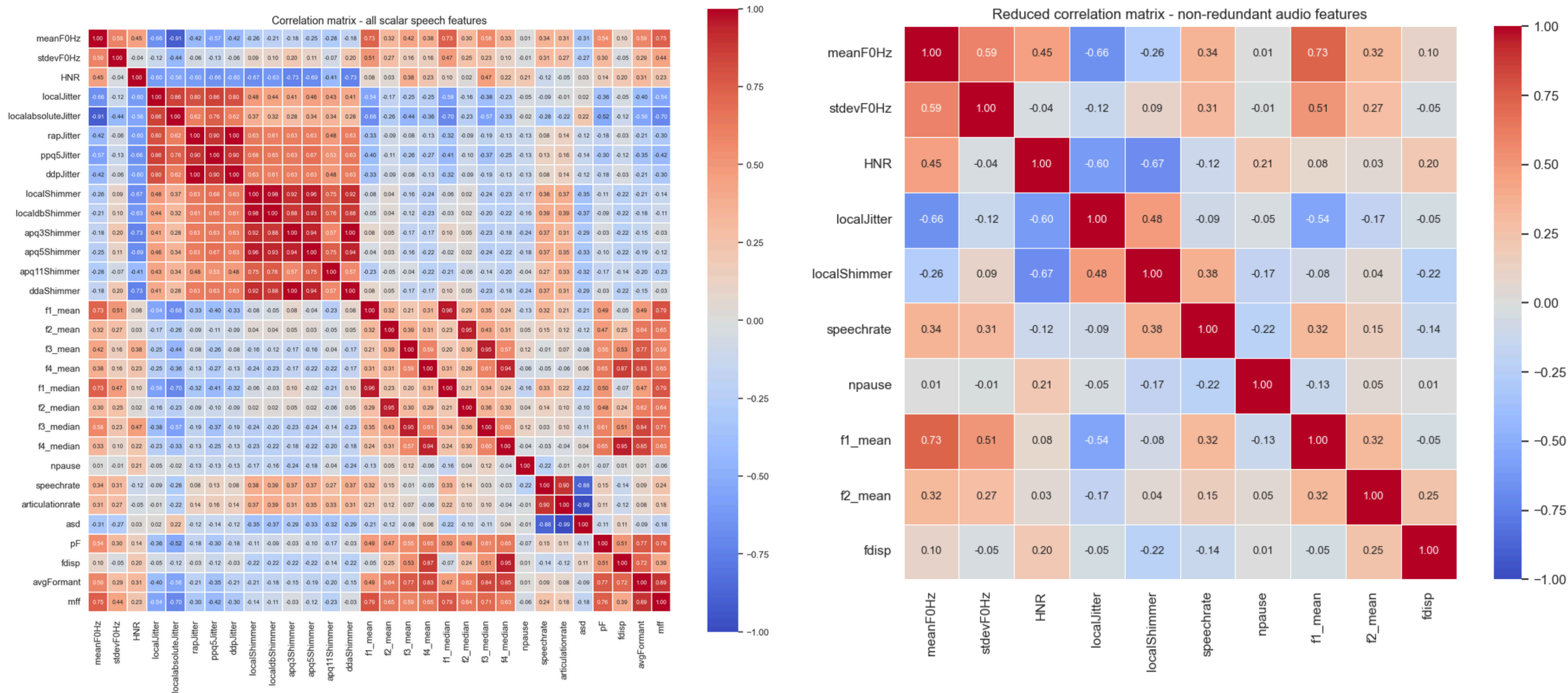
Audio and Speech Features

Audio Transcript Quality (Debates & Newscast's) - Random characters & Non Portuguese Text



Audio and Speech Features

Audio Features - Correlation Matrix



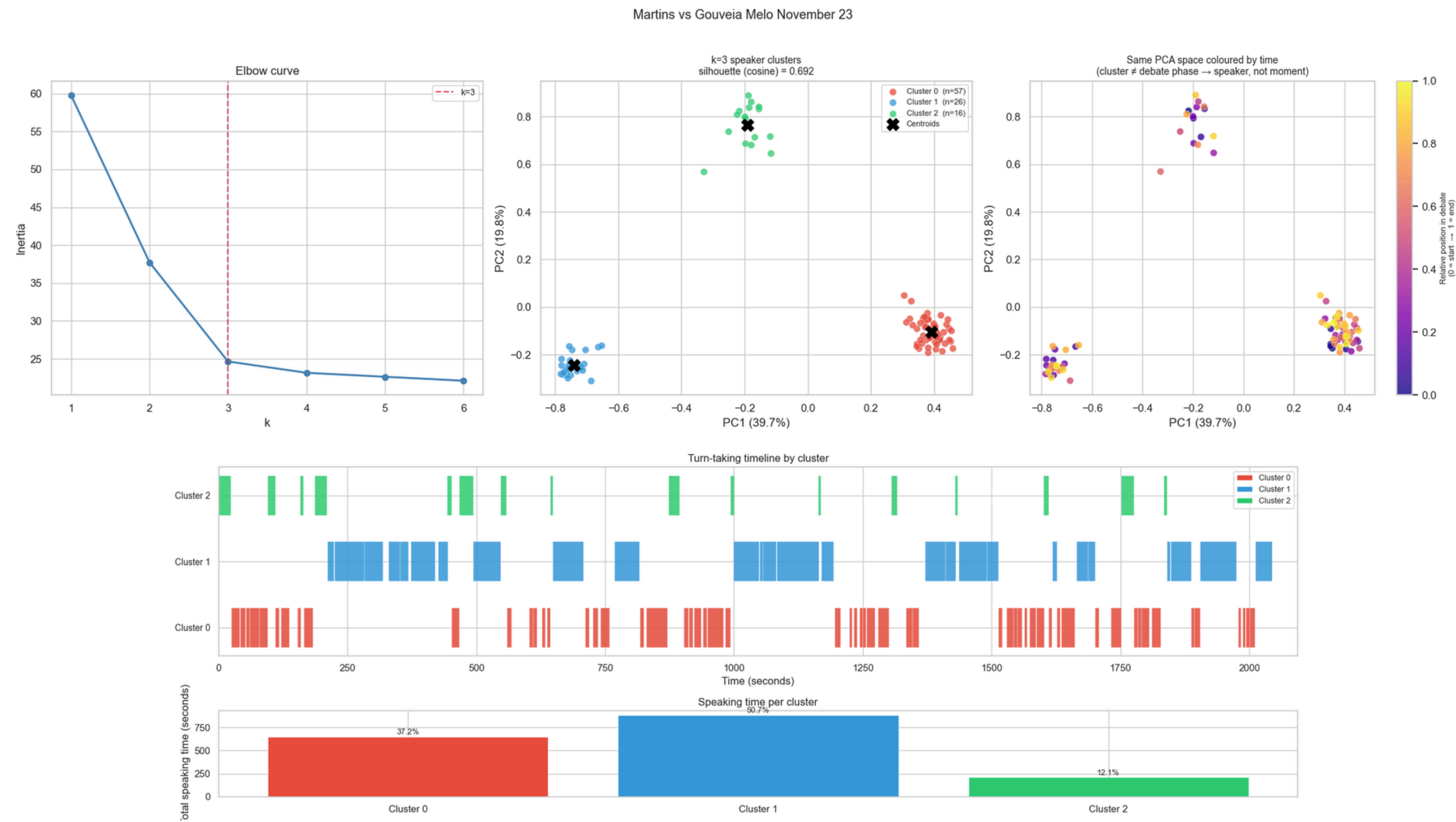
Audio and Speech Features

Audio Transcripts — Debate's Word Cloud



Audio and Speech Features

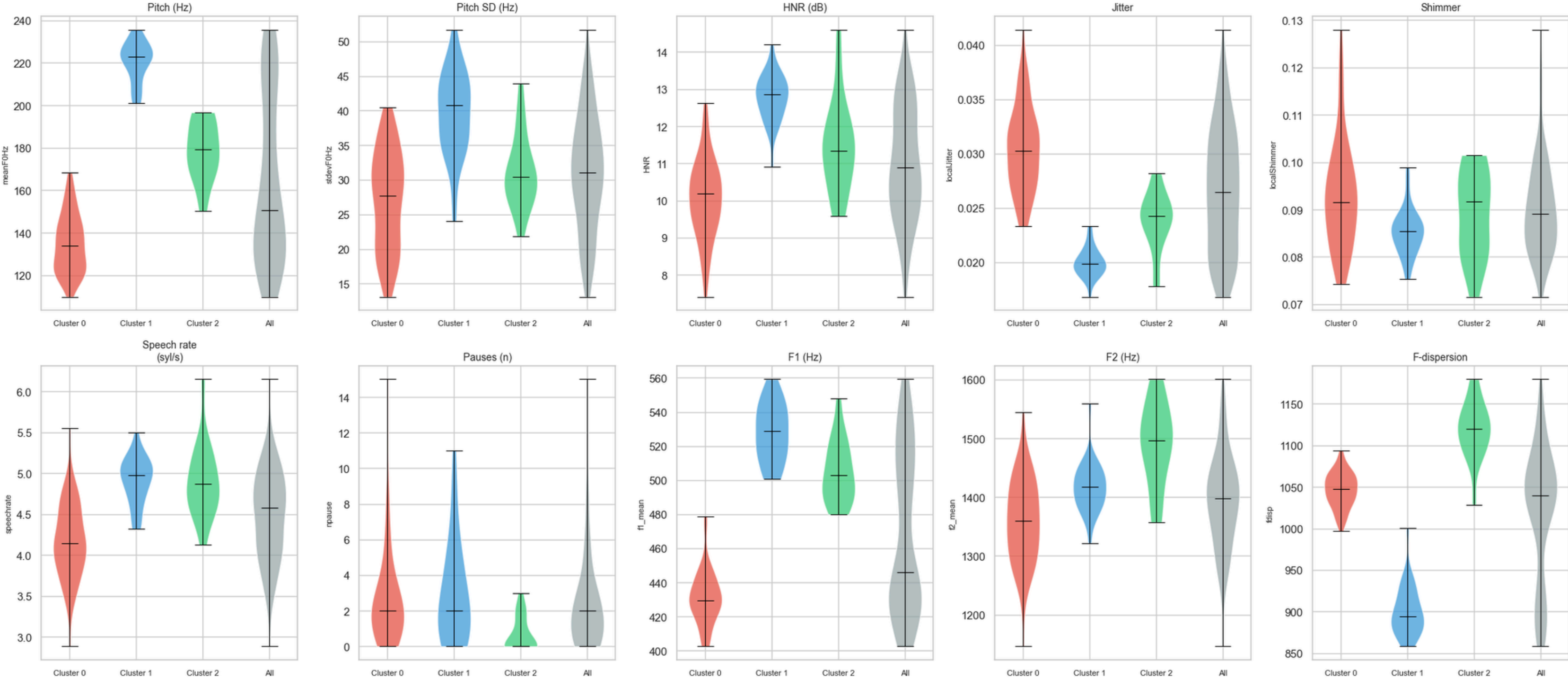
Speaker embeddings — PCA & KMeans clustering



Audio and Speech Features

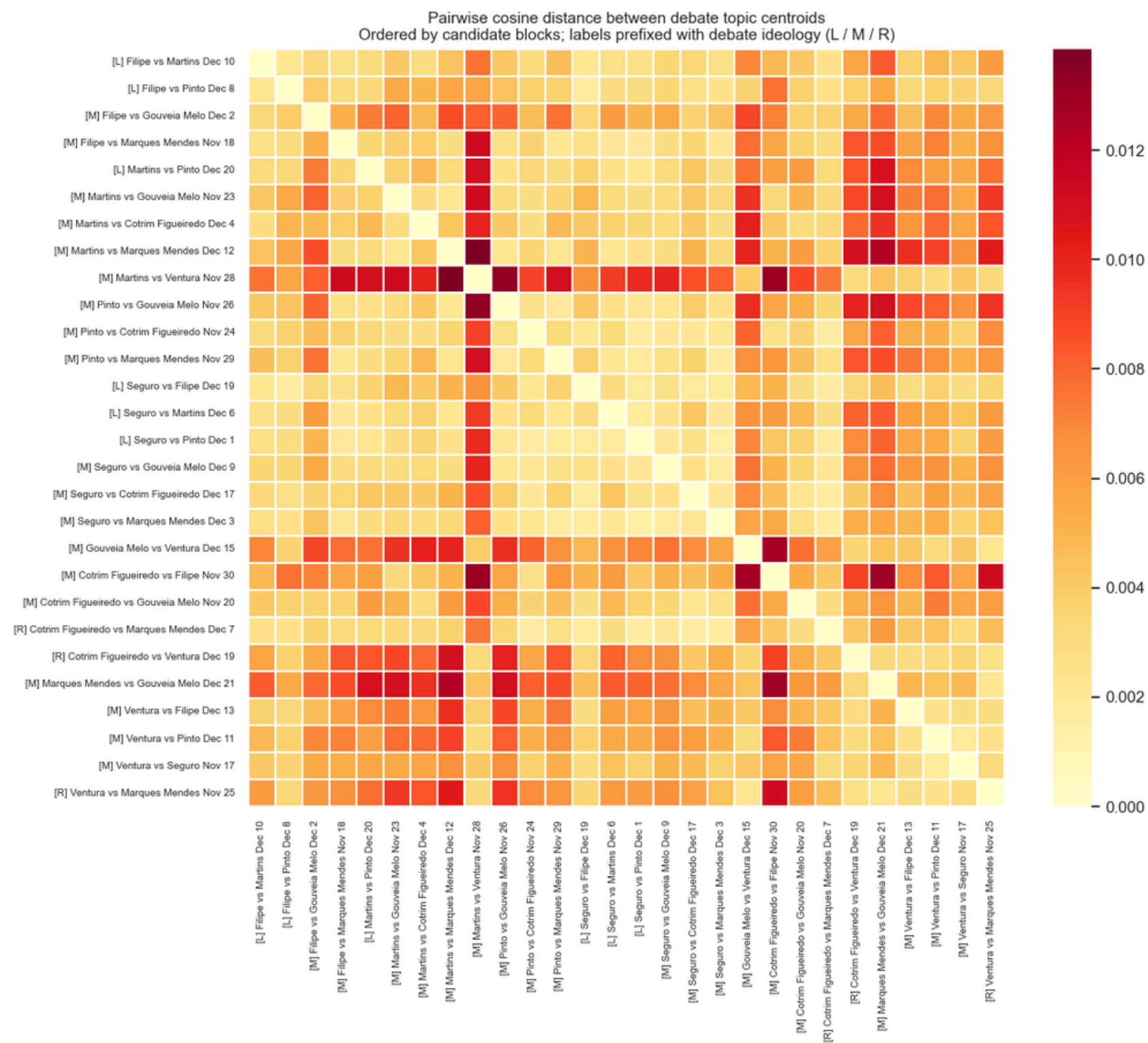
Debate audio & speech — distributions per speaker segment

Audio feature distributions by speaker cluster — Martins vs Gouveia Melo November 23



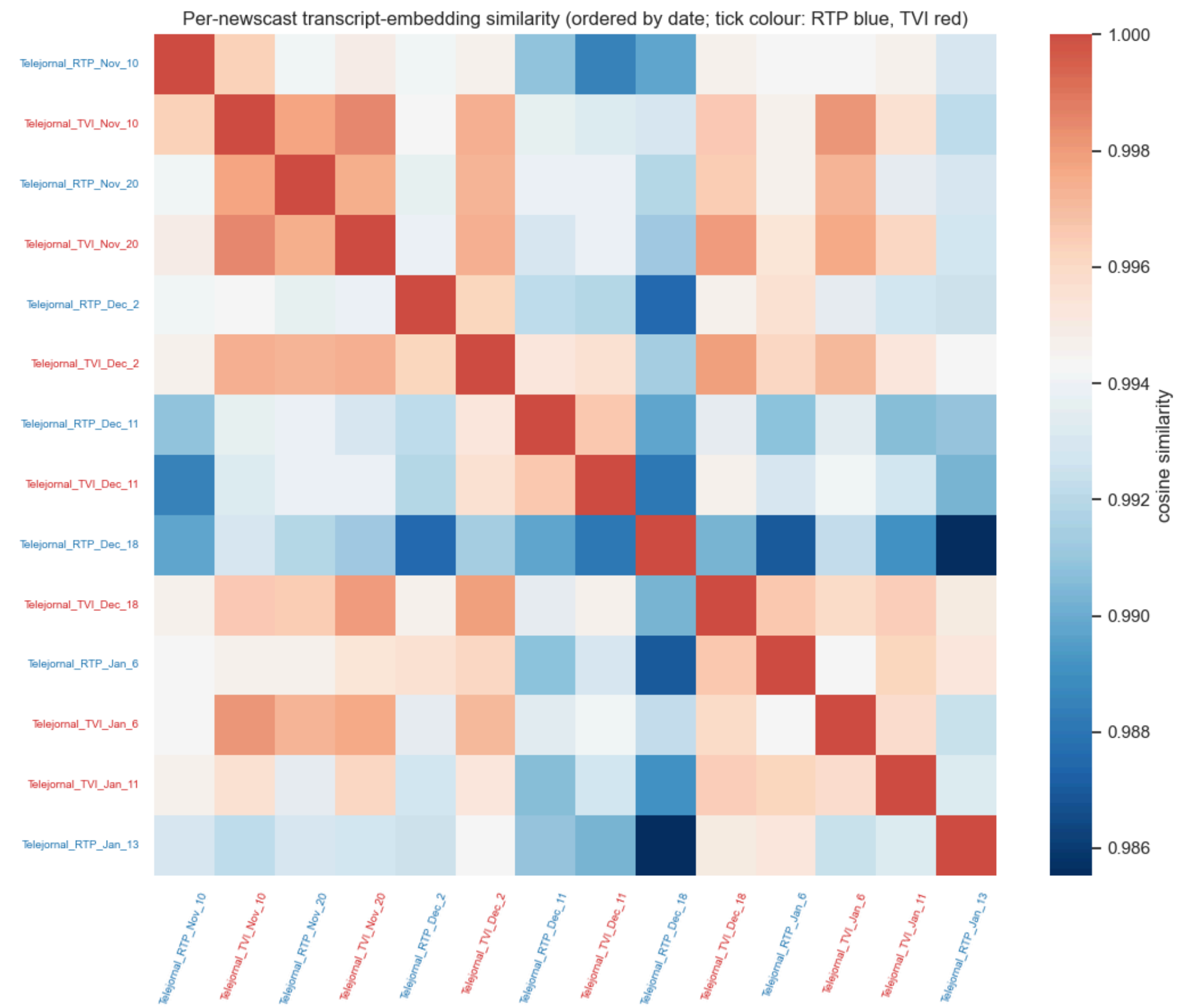
Audio and Speech Features

Transcript embeddings — Correlation Matrix & Word Cloud's



Audio and Speech Features

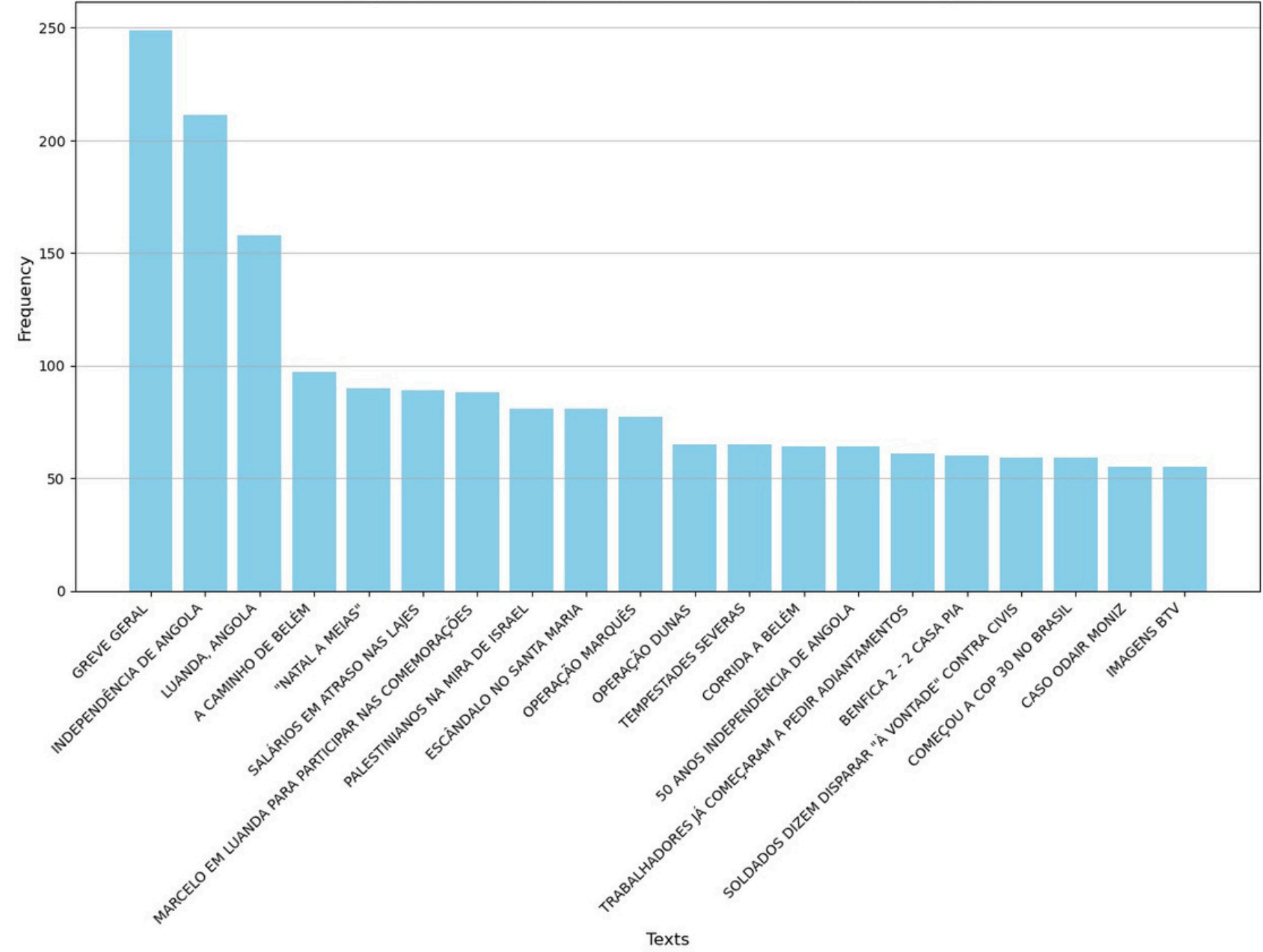
Newscast transcript embeddings — Correlation between channels



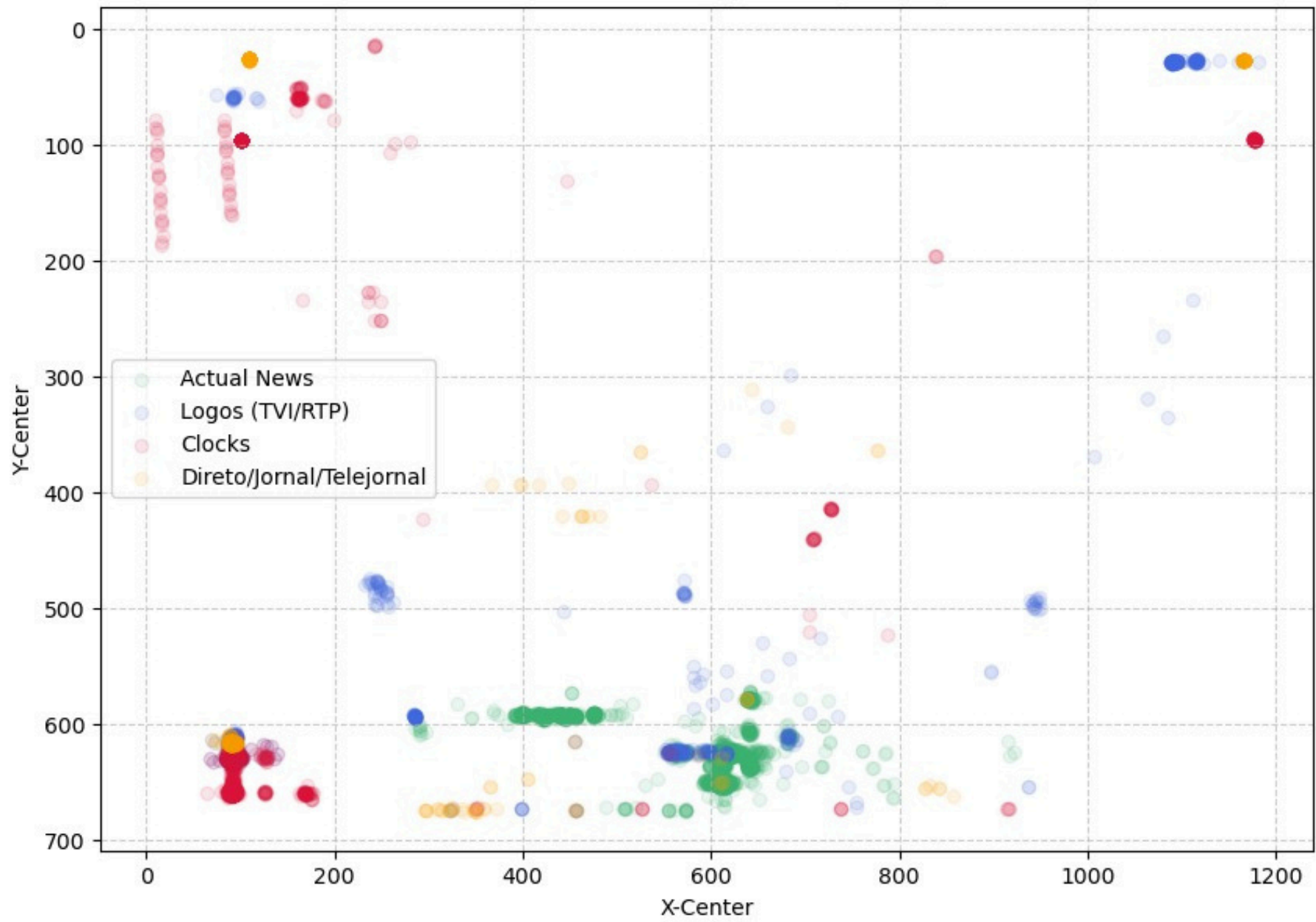
Visual and Text Features

Newcast OCR: News frequency and text location

Top 20 Most Common OCR Texts (Clean Clock and Logos)



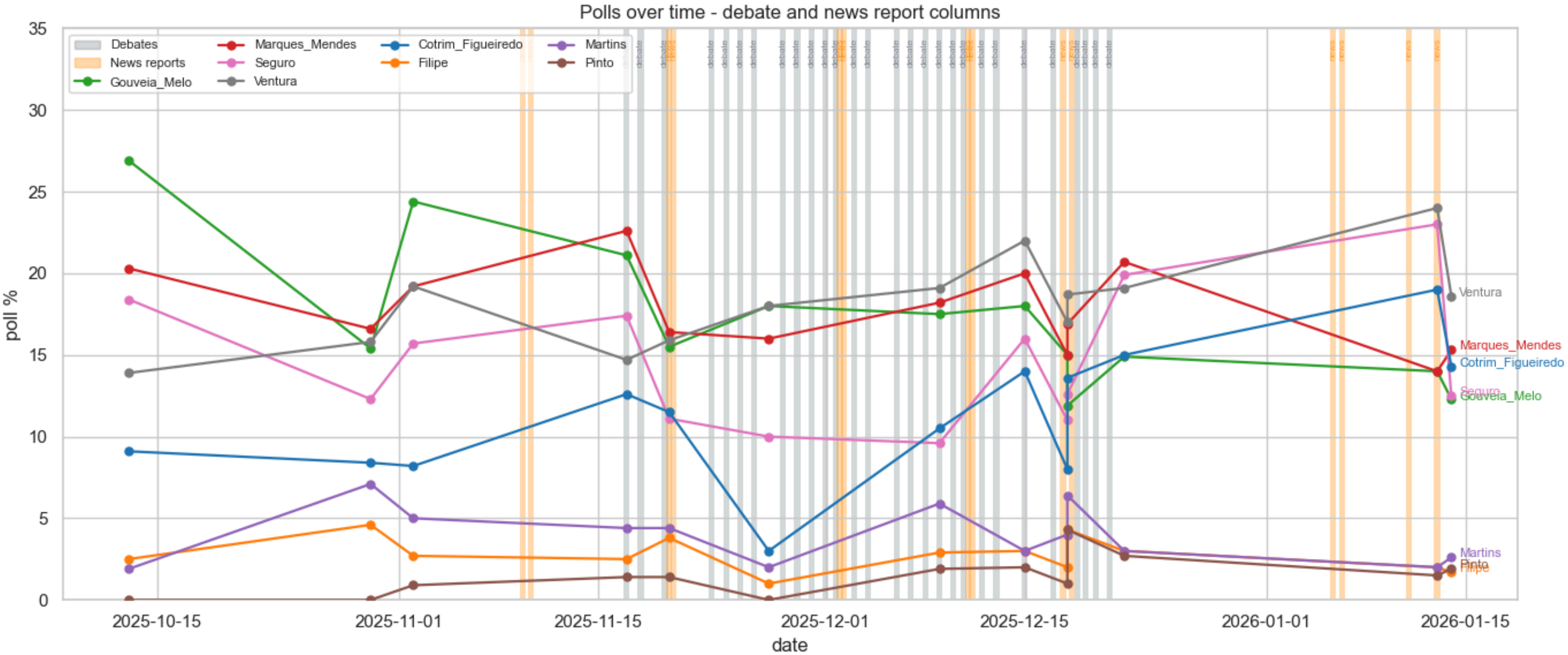
Centroid of Bounding Boxes (filtered)



Example: TVI - November 10

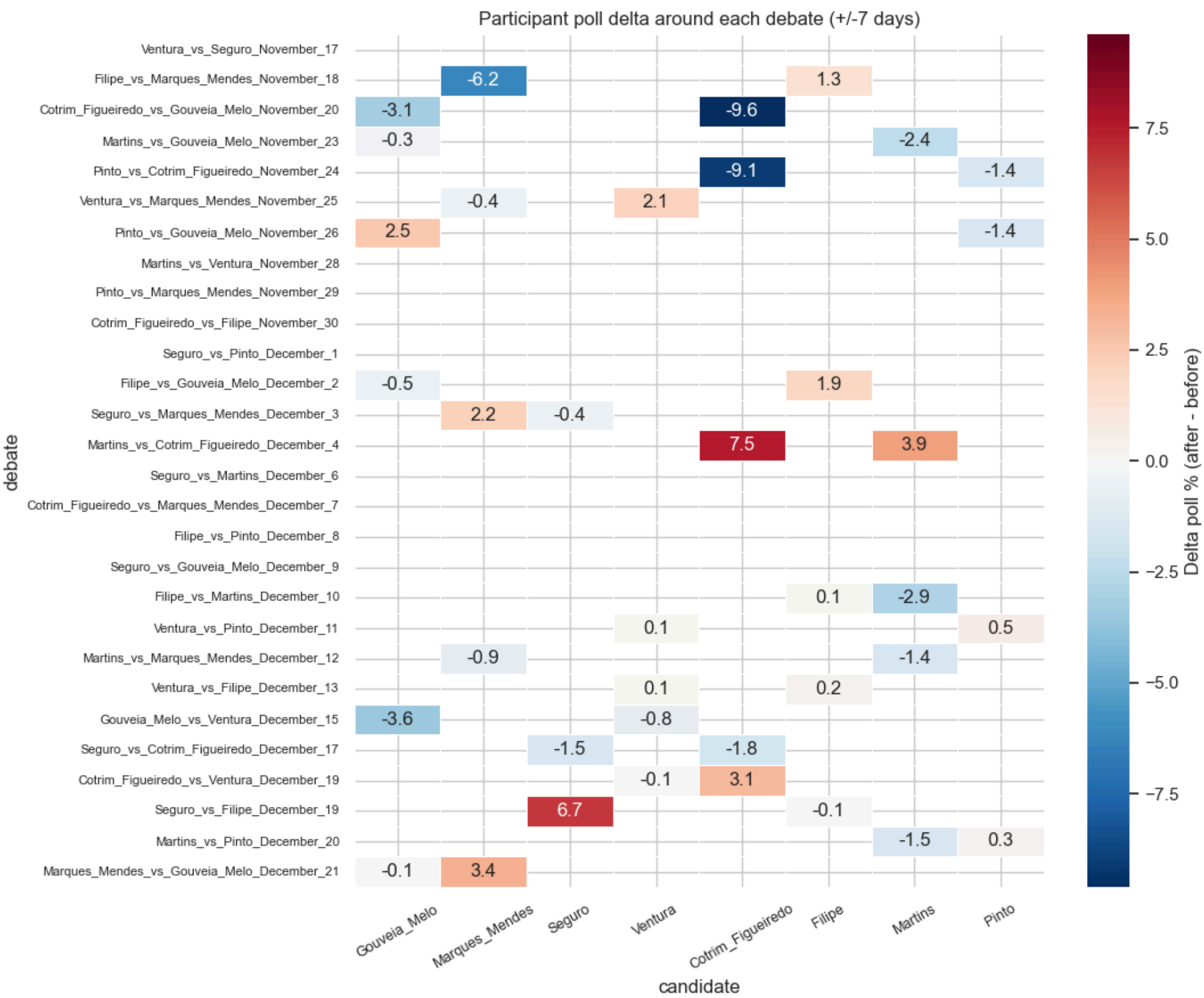
Text Features

Poll Results vs Debate Timeline



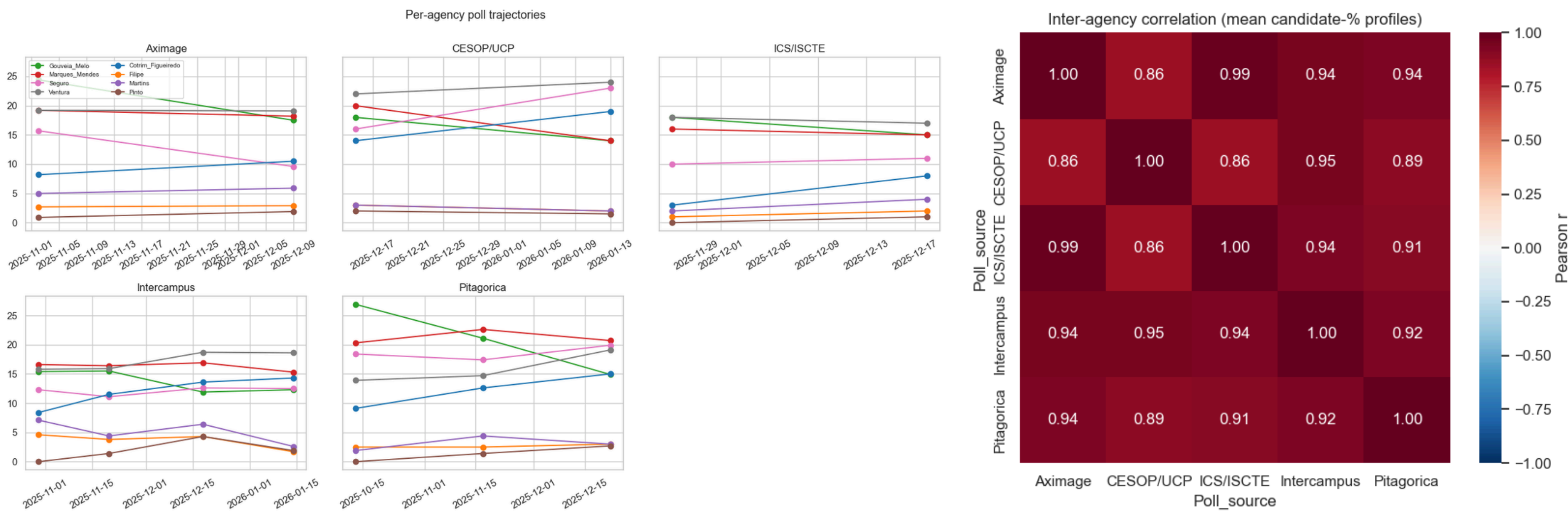
Text Features

Poll Results vs Debate Timeline



Text Features

Poll Results vs Debate Timeline



Questions